KEMENTERIAN PENDIDIKAN TINGGI

PROGRAM PENINGKATAN SKIL USAHAWAN DIGITAL



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Digital Forensics and the Investigator



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UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

Menginovasi Penyelesaian Lestari

DFBOOK DIGITAL FORENSICS · LESSON 1.0

Digital Forensics and the Investigator

Lesson overview: the field, the authority and the method



LESSON 1.0 · OVERVIEW

The Field and the Method

What the work is, who may allow it, and how it is done

BY THE END YOU WILL BE ABLE TO

Explain what digital forensics is

And how it differs from data recovery and incident response

Choose the authority a case needs

A legal power for the police; policy and a requester in a company

Follow a systematic approach

From a first hypothesis to a planned and checked case

Keep notes and conduct sound

Objective, confidential, and recorded as you work

LESSON 1.0 · ROUTE

Three Short Sub-Lessons

One idea and one desk activity in each

LESSON 1.1

What Digital Forensics Is

The field, the work around it, and evidence for and against a suspect.

ACTIVITY **Scenario sort**

LESSON 1.2

Public and Private Investigations

Who may investigate, and what gives them the right to.

ACTIVITY **Who may authorise?**

LESSON 1.3

A Systematic Approach to a Case

Five steps, a planned approach, and your first notes.

ACTIVITY **First hypothesis**

From Lesson 1.2 onwards, the desk activities work on one fictional case: Teratai Supplies Sdn Bhd.

LESSON 1.0 · THE CASE

The Teratai Supplies Case

A fictional company, one problem, and five likely sources of evidence

THE COMPANY

Teratai Supplies Sdn Bhd

THE PROBLEM

Auditors find payments for materials that were never delivered. The accounts payable executive approved them.

LIKELY SOURCES OF EVIDENCE

- **Company laptop**
- **USB drive in the desk**
- **Mobile phone**
- **Email accounts**
- **Cloud accounting export**

The question: an honest error or a fraud? Let the digital evidence show which.

UP NEXT

What Digital Forensics Is

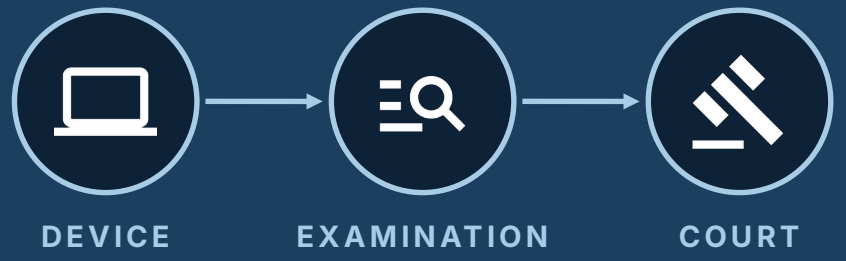
We start by defining the field, then set it apart from the work around it.

- **Defining the field**
- **Forensics and related work**
- **Evidence for and against**

DFBOOK DIGITAL FORENSICS · LESSON 1.1

What Digital Forensics Is

The field, the work around it, and evidence for and against



IN THIS LESSON

1. Defining the field
2. Forensics and related work
3. Evidence for and against

LESSON 1.1 · OVERVIEW

What Digital Forensics Is

Digital forensics is more than getting files back. It applies computer science and investigative procedure, so that what an examiner finds can be used as evidence.

BY THE END YOU WILL BE ABLE TO

1. **Define digital forensics and its requirements**
2. **Tell it apart from recovery and response**
3. **Explain inculpatory and exculpatory evidence**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Three milestones in how digital evidence came to be handled

1984

A dedicated team

The FBI forms its Computer Analysis and Response Team (CART).

2007

A field of specialties

Ken Zatyko describes specialties such as computer, network and video forensics.

2012

An international standard

ISO/IEC 27037 sets guidelines for collecting and preserving digital evidence.

WHERE DIGITAL EVIDENCE IS FOUND TODAY

- **Computers**
- **Mobile phones**
- **USB drives**
- **Cloud storage**
- **Internet of Things**

Internet of Things (IoT): everyday objects linked to the Internet through sensors and chips, such as cars and TVs.

INTRODUCTION · ROADMAP

Your Route Through Lesson 1.1

Each part answers one question, in order

1

PART 1

Defining the field

What is digital forensics, and what must it include?

2

PART 2

Forensics and related work

How does it differ from data recovery and incident response?

3

PART 3

Evidence for and against

Why look for evidence that clears a suspect too?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Defining the Field

What digital forensics is, and what its results rest on

01

IN THIS PART YOU WILL LEARN TO

- Define digital forensics
- Name the requirements a forensic result rests on

KEY TERMS

Digital forensics **Hash value** **Repeatability**

PART 1 · DEFINITION

Defining Digital Forensics

The definition used by NIST, the US National Institute of Standards and Technology

> **“The application of computer science and investigative procedures involving the examination of digital evidence”, with seven requirements:**

1 Search authority

Legal permission to search (Lesson 1.2).

2 Chain of custody

Who held the evidence, and when (Lesson 2).

3 Validation with mathematics

A hash value, calculated from the data, confirms a copy is exact.

4 Validated tools

Tools tested to give correct results.

5 Repeatability

Another examiner repeats the steps and gets the same result.

6 Reporting

Findings written up clearly (Lesson 8).

7 Expert testimony

Explaining findings in court, if needed.

Each one is taught later in the module.

PART 1 · THE WORK

What Examiners Do

Three stages of the work, with the law applied throughout

STAGE 1 → STAGE 2 → STAGE 3

Collect securely

Gather the data so that nothing is changed or lost.

Examine

Work out details such as where data came from and what it holds.

Present

Explain the digital information to a court or an inquiry.

Throughout: apply the relevant laws to how digital devices are used.

Digital evidence: information that can help prove or disprove a fact, stored or sent in binary form.

Like an archaeologist, an examiner digs out information that may already be on a drive but is hard to find or read.

PART 1 · STANDARDS

Standards and the Law

The international standard, and the law where you work

INTERNATIONAL STANDARD

ISO/IEC 27037 (2012)

Guidelines for identifying, collecting, acquiring and preserving digital evidence.

WHY IT MATTERS

Shared methods let countries share evidence.

THE LAW WHERE YOU WORK

Malaysian law, not US law

The textbook cites US rules. In Malaysia, the Computer Crimes Act 1997 defines data, and the Evidence Act 1950 governs evidence in court.

BEFORE YOU RELY ON IT

Check the current wording with a legal adviser.

Investigator and examiner: the textbook uses both words for the same role. In some organisations they are different people.

PART 2 OF 3

Forensics and Related Work

How digital forensics differs from the work around it

02

IN THIS PART YOU WILL LEARN TO

- Tell digital forensics from data recovery
- Place it within the wider security team

KEY TERMS

Data recovery **Incident response** **Network forensics**

PART 2 · COMPARISON

The Wider Field

Four activities, their main aim, and whether results must stand as evidence

Activity	Main aim	Results used as evidence?
Digital forensics	Evidence for civil, criminal or administrative cases	**Yes, always**
Incident response	Protect the organisation's assets and contain an attack	Not the main aim
E-discovery	Find electronic records for a civil lawsuit	Yes, in civil cases
Research	Develop new methods and tools	No

What sets digital forensics apart: its results must stand up as evidence.

PART 2 · CONTRAST

Forensics and Data Recovery

Data recovery compared with digital forensics

Data recovery

- Gets back data deleted by mistake, or lost in a crash or power surge
- You usually know what you are looking for
- The owner just wants the data back

Digital forensics

- Recovers data that users have hidden or deleted
- What the device holds is unknown at the start
- What is recovered must be valid as evidence

Data recovery can involve deliberate deletion too. The difference is the goal: results that are valid as evidence.

PART 2 · SPECIALTIES

Computer and Network Forensics

Two specialties, compared on the same three points

Computer forensics

LOOKS AT

Data stored on a drive or other storage media

FINDS

Files that are there, hidden or deleted

TYPICAL SOURCE

A hard drive or USB drive

Network forensics

LOOKS AT

Activity across a network

FINDS

How attackers got in; files copied or changed

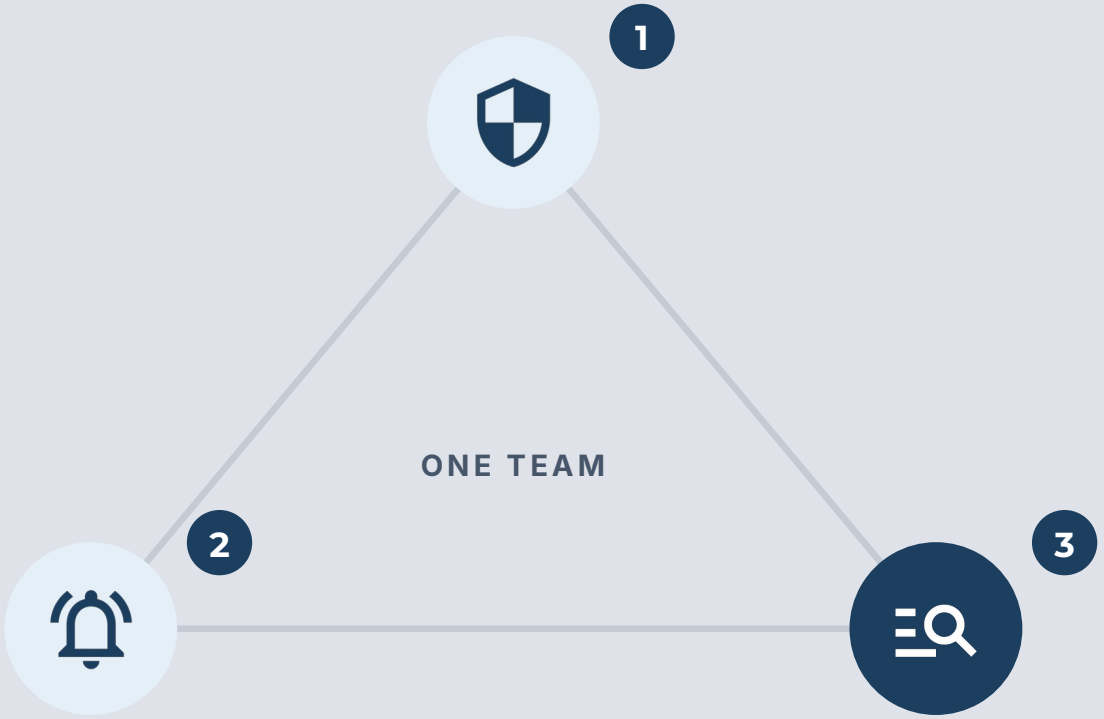
TYPICAL SOURCE

Log files: logons, websites visited, locations

PART 2 · STRUCTURE

The Investigations Triad

Three groups that together secure an organisation's computers and networks



1 Vulnerability and risk assessment

Tests the security of systems and networks; includes penetration testers.

2 Intrusion detection and incident response

Spots attacks with automated tools and firewall logs, then tracks and stops intruders.

3 Digital investigations

Images and analyses drives, and has final responsibility for resolving a case.

In a small company, one group may do all three, or the work is contracted out.

PART 3 OF 3

Evidence For and Against

Why examiners look for evidence on both sides

03

IN THIS PART YOU WILL LEARN TO

- Explain why digital evidence starts as an unknown
- Tell inculpatory from exculpatory evidence

KEY TERMS

Inculpatory **Exculpatory**

PART 3 · CONTRAST

Physical and Digital Evidence

What can be identified on sight, and what cannot

Physical evidence

Blood, hair or a bullet can be identified as soon as it is found.

Digital evidence

What a laptop or phone holds is unknown until it is examined.

WHAT THIS MEANS FOR THE EXAMINER

1. Examiners often start without knowing if a device holds evidence at all.
2. They search the storage media and piece together the data they find.
3. Software handles most cases; damaged media may need costly lab equipment.

Example of costly equipment: electron microscopes, used on drives damaged or reformatted on purpose.

PART 3 · DEFINITION

Inculpatory and Exculpatory Evidence

Two kinds of evidence an examiner looks for in every case

INCULPATORY

Points towards guilt

Evidence that tends to show the suspect did it. In criminal cases it is called incriminating.



EXCULPATORY

Points away from guilt

Evidence that tends to clear the suspect, pointing towards innocence.

Search for both

Every device is searched with both kinds in mind.

Record both

Findings that clear a suspect are written up too.

The evidence decides the outcome, not the examiner.

PART 3 · WORKED EXAMPLE

Four Possible Findings

Teratai Supplies: illustrative findings for practice, not facts of the case

THE CASE

Payments for undelivered materials were approved under the accounts payable executive's login.

QUESTION

Which findings point towards the executive, and which point away?

1	Approvals sent from the executive's email account	↗ INCULPATORY
2	Fake supplier invoices in the executive's folders	↗ INCULPATORY
3	The executive was on leave for two of the approvals	↘ EXCULPATORY
4	A colleague logged in with the executive's password	↘ EXCULPATORY

ANSWER

Two point towards the executive and two point away. All four go in the report.

PART 3 · WORKED EXAMPLE

Sorting Requests by Purpose

Two questions sort most requests that reach an examiner

1 **Could it lead to court, discipline or a dispute?**

YES → **Digital forensics** Auditors suspect fake supplier invoices

NO ↓

2 **Is the aim to stop an attack and protect systems?**

YES → **Incident response** Malware is spreading on the payroll server

NO ↓

Data recovery A budget file deleted by mistake

Tip: if unsure, treat it as forensics; a civil matter can turn criminal.

Incident response becomes forensic work when evidence is collected for legal action against an intruder.

LESSON 1.1 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Defining the field

Computer science and investigative procedure applied to digital evidence, resting on seven requirements.

2

Forensics and related work

Data recovery and incident response use similar skills; forensics adds the need for valid evidence.

3

Evidence for and against

Device contents start unknown. Examiners look for inculpatory and exculpatory evidence alike.

Every result must hold up as evidence, whichever way it points.

UP NEXT

Public and Private Investigations

Next: who may investigate, and what gives them the right to.

- **Two kinds of investigation**
- **Authority in a company**
- **Company cases in practice**

DFBOOK DIGITAL FORENSICS · LESSON 1.2

Public and Private Investigations

Who may investigate, and what gives them the right to

IN THIS LESSON

1. Two kinds of investigation
2. Authority in a company
3. Company cases in practice



LESSON 1.2 · OVERVIEW

Public and Private Investigations

Before any device is examined, someone must have the right to examine it. That right comes from the law in a public case, and from company policy in a private one.

BY THE END YOU WILL BE ABLE TO

1. **Tell public from private investigations**
2. **Name the authority each one needs**
3. **Spot when a company case turns criminal**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Why every investigation is handled as if it could reach a criminal court

> **“If you follow good forensics procedures, the evidence found in your examinations can make the transition between civil and criminal cases.”**
>
> Guide to Computer Forensics and Investigations, on private-sector cases

Cases can change

A company inquiry can turn into a criminal case.

Authority comes first

Without the right authority, evidence may be challenged or excluded.

One standard for all

Apply the same rules of evidence to every case you handle.

Civil and criminal cases: a civil case is a dispute between parties, such as a company and an employee; a criminal case is brought by the state.

INTRODUCTION · ROADMAP

Your Route Through Lesson 1.2

Each part answers one question, in order

1

PART 1

Two kinds of investigation

Who investigates, about what, and under what rules?

2

PART 2

Authority in a company

What gives a company the right to investigate?

3

PART 3

Company cases in practice

How are common company cases handled, and when do they turn criminal?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Two Kinds of Investigation

Public and private cases, and the rules each one follows

01

IN THIS PART YOU WILL LEARN TO

- Compare public and private investigations
- Follow the stages of a criminal case

KEY TERMS

Allegation **Affidavit** **Search warrant**

PART 1 · COMPARISON

Public and Private Investigations

The two kinds compared on the same four points

	Public sector	Private sector
Who investigates	Police and other government agencies	Companies, their lawyers and investigators
Typical cases	Crimes such as fraud, theft or harassment	Policy breaches, disputes, white-collar crime
Limited by	Search-and-seizure law	Company policy and privacy law
Main aim	A case that can be prosecuted	Stop the violation and limit the loss

A private case can become criminal, and a criminal case can lead to a civil one.

PART 1 · SEQUENCE

Stages of a Criminal Case

Three stages, then the examiner's role (not the five steps of Lesson 1.3)

STAGE 1 → STAGE 2 → STAGE 3

Complaint

A witness or victim makes an allegation; an officer writes a report.

Investigation

A specialist collects evidence and, with the prosecutor, builds a case.

Prosecution

If the evidence is strong enough, the case goes to trial.

THE EXAMINER'S ROLE IN A POLICE CASE

Assess the scope → **Decide resources and specialists** → **Gather them** → **Delegate, collect and process** → **Present a report**

Allegation: a statement, made to the police, that a crime has been committed.

PART 1 · LEGAL PROCESS

Search Warrants and Roles

The textbook's US route, with Malaysian powers below



Affidavit

A sworn statement of facts that justify a search

Search warrant

Approved and signed by a judge

Collection

A first responder (DEFR) collects what it covers

Analysis

A specialist (DES) analyses the data

Court

Evidence is presented at a hearing or trial

IN MALAYSIA · CRIMINAL PROCEDURE CODE

1. **Section 116B:** Access to computerised data during a search, including passwords and decryption.
2. **Section 116C:** Interception of communications needs the Public Prosecutor's authorisation.

The textbook describes US procedure. Confirm current Malaysian provisions with a legal adviser.

PART 2 OF 3

Authority in a Company

What gives a company the right to investigate its own systems

02

IN THIS PART YOU WILL LEARN TO

- Explain what company policies and banners do
- Name who may request an investigation

KEY TERMS

Acceptable use **Warning banner** **BYOD**

PART 2 · STRUCTURE

Company Policies

Three things a policy sets out, and the risk of having none



ACCEPTABLE USE POLICY

Rules for company computers

Staff sign an agreement to follow them.

LINE OF AUTHORITY

Who may act

Who may start an inquiry, hold evidence and see it.

DUE PROCESS

Fair treatment

A promise to treat every employee fairly under the law.

WITHOUT CLEAR POLICIES

- Investigators may lack authority
- Staff may assume their use is private
- The company risks lawsuits

Keep policies in line with current law, and train staff on them regularly.

PART 2 · EXHIBIT

Warning Banners

A notice shown at log-on, and what each part of it does

COMPANY NETWORK · NOTICE

1. Access is for authorised users and official business only.
2. Systems may be monitored at any time.
3. Using this system means you consent to monitoring.
4. Users have no expectation of privacy.

OK

MOCK · ILLUSTRATIVE

1 Restricts use

Authorised people and company business only.

2 Claims monitoring rights

The company may inspect systems at any time.

3 Asks for consent

Logging on counts as agreeing to be monitored.

4 Removes privacy

Staff cannot claim their use was private.

A clear banner lets a company inspect its own equipment without a warrant.

PART 2 · ROLES

The Authorised Requester

Who may ask for an investigation, and how everyone else asks

GROUPS THAT MAY REQUEST AN INVESTIGATION

- Corporate security investigations team
- Corporate ethics office
- Equal employment opportunity office
- Internal audit department
- General counsel or legal department

EVERYONE ELSE

Any other department

↓

Corporate security team

↓

Investigation starts

Fewer requesters means fewer false allegations, and keeps inquiry separate from discipline.

PART 2 · CATEGORY MAP

Personal and Company Property

Who owns the device, and whether it connects to the company network

	COMPANY-OWNED	PERSONAL
CONNECTED	**Company rules apply** Covered by policy and the banner.	**Unclear** Data flows both ways when it syncs.
NOT CONNECTED	**Company rules apply** Even when used away from the office.	**Personal property** Needs consent or a legal power.

QUESTIONS THE TEXTBOOK RAISES

1. Does data synced from a personal tablet belong to the company?
2. If the company gave the tablet as a gift, can it still claim it?
3. What does the company's BYOD policy say?

BYOD (bring your own device): staff use personal devices on the company network.

PART 3 OF 3

Company Cases in Practice

Common company cases, and when they turn criminal

03

IN THIS PART YOU WILL LEARN TO

- Describe common company cases
- Recognise when to involve the police

KEY TERMS

Email abuse **Internet abuse** **Interrogation**

PART 3 · CATEGORIES

Common Company Cases

Three situations compared. Consult legal and HR teams before starting

Misuse of IT

WHAT IT IS

Breaking company rules with company IT

EXAMPLE

Running a side business on company software

RISK

Lost time and company resources

Email abuse

WHAT IT IS

Misusing the company email system

EXAMPLE

Offensive, harassing or threatening messages

RISK

A hostile workplace and lawsuits

Internet abuse

WHAT IT IS

Misusing company Internet access

EXAMPLE

Browsing all day, or viewing obscene content

RISK

Illegal content is a crime: call the police

PART 3 · PROCEDURE

Internet and Email Abuse Cases

What you need, and the key check. Privacy law can override company policy

Internet abuse

YOU NEED

- ✓ Proxy server logs (websites visited)
- ✓ The computer's IP (network) address
- ✓ The computer's drive
- ✓ A forensics analysis tool

KEY CHECK

Compare web history on the drive with the logs. If neither supports it, report the allegation as unproven.

Email abuse

YOU NEED

- ✓ The email, with its header (routing details)
- ✓ Email server logs
- ✓ The server, or the email file on the computer
- ✓ A forensics analysis tool

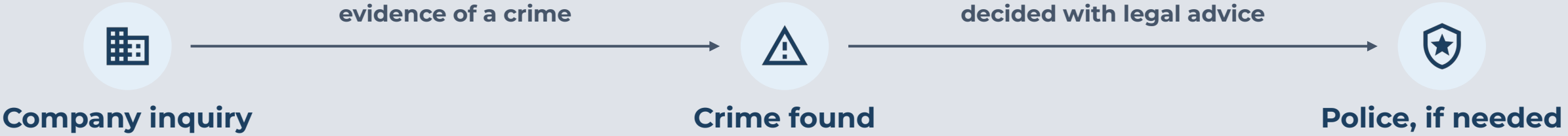
KEY CHECK

Examine the header of every message of interest.

PART 3 · ROUTE

When a Company Case Turns Criminal

What changes when a crime comes to light



1 Illegal images

Stop, and notify law enforcement.

2 A victim outside the company

You may have a legal and moral duty to report.

3 Working for the police

You become their agent: they cannot ask you to get what needs a warrant.

Any company case can become criminal, so handle all evidence with the highest level of security and accountability.

PART 3 · CATEGORIES

Three Special Situations

Each changes who directs the work and how it is done

Working for a lawyer

The lawyer directs the work.

- Start only after the lawyer's written memo
- Keep every finding confidential
- Mark documents as privileged legal communication

Industrial espionage

Treat it as criminal from the start.

- Plan discreetly with lawyers and management
- Prove the allegation before going to the police
- Use a specialist team; watch the scope

Interviews

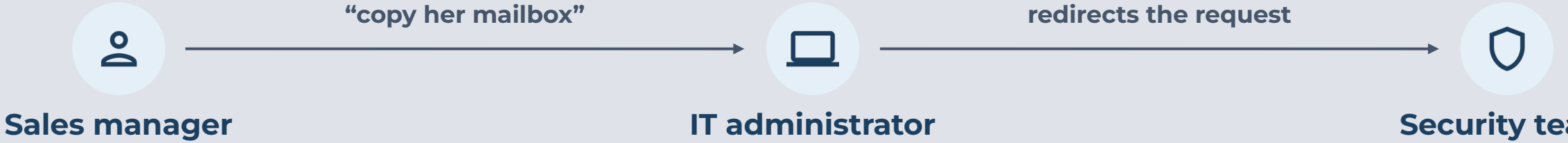
You support the interviewer.

- Suggest questions, and the answers to expect
- An interview gathers facts; an interrogation seeks a confession
- Be patient and persistent

PART 3 · WORKED EXAMPLE

The Request from Sales

Teratai Supplies, illustrative: a manager asks IT to copy a mailbox



1 Is the manager an authorised requester?

No. The request must go through the corporate security team.

2 Could this be a crime?

Selling price lists to a rival is industrial espionage: handle it as criminal.

ANSWER

Send it through security, under company policy, and treat the evidence to criminal standards.

PART 3 · WORKED EXAMPLE

Devices and Accounts at Teratai

Which items the company may examine under its own policy

THE CASE

The company wants the accounts payable executive's devices and accounts examined.

QUESTION

Which items does company policy cover?

1	**Company laptop** Company property, under policy and the banner	COVERED
2	**Company email accounts** A company system, copied by its administrator	COVERED
3	**USB drive in the desk** Owner unclear: confirm before examining	CHECK
4	**Mobile phone** If personal: consent, BYOD policy or the police	CHECK
5	**Cloud accounting export** Company records; cloud evidence is Lesson 7	COVERED

ANSWER

Policy covers company systems. For personal or unclear items, get consent or a legal power.

LESSON 1.2 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

Two kinds of investigation

Public cases follow the law on search and seizure; private cases follow company policy.

Authority in a company

Policies, banners and an authorised requester give a company the right to investigate.

Company cases in practice

Misuse, email and Internet abuse are common. Some crimes must go to the police.

Confirm your authority before you touch a device, and handle every case as if it may reach court.

UP NEXT

A Systematic Approach to a Case

Next: the steps of an investigation, a planned approach, and your first notes.

- **Five steps of an investigation**
- **A systematic plan**
- **Conduct and notes**

DFBOOK DIGITAL FORENSICS · LESSON 1.3

A Systematic Approach to a Case

Five steps, a planned approach, and your first notes

IN THIS LESSON

1. Five steps of an investigation
2. A systematic plan
3. Conduct and notes



LESSON 1.3 · OVERVIEW

A Systematic Approach to a Case

Good investigations are not improvised. A method, a written plan and careful notes make your findings reliable, and let someone else check them.

BY THE END YOU WILL BE ABLE TO

1. **Put the five steps of an investigation in order**
2. **State a first hypothesis for a case**
3. **Plan a case step by step**
4. **Keep notes and conduct that hold up**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

A method protects both the evidence and the examiner

5

Steps in a typical case

From a first hypothesis to a final report.

12

Tasks in a systematic plan

So that nothing is overlooked.

1

First rule for every case

Preserve the evidence: no tampering, no contamination.

WHERE YOUR WORK IS CHECKED

In court | **At a private inquiry** | **By the other side's expert**

Case: a collection of evidence that can be offered in court or at a private-sector inquiry.

INTRODUCTION · ROADMAP

Your Route Through Lesson 1.3

Each part answers one question, in order

1

PART 1

Five steps of an investigation

What steps does every investigation follow?

2

PART 2

A systematic plan

How do you plan a case so that nothing is missed?

3

PART 3

Conduct and notes

How do you keep your work credible?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Five Steps of an Investigation

The cycle every investigation follows, and how it ends

01

IN THIS PART YOU WILL LEARN TO

- Put the five steps in order
- Explain why the first four steps repeat

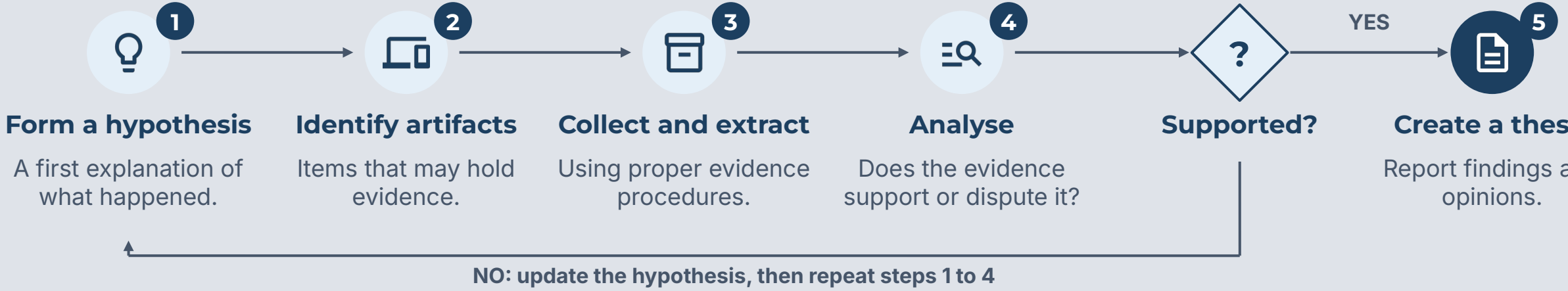
KEY TERMS

Hypothesis **Artifact** **Thesis**

PART 1 · SEQUENCE

Five Steps of an Investigation

Steps 1 to 4 repeat until the evidence and the hypothesis agree



Hypothesis: a proposed explanation of what happened, to be tested against the evidence.

Artifact: an item that may hold evidence, such as a laptop, a phone or a server.

Thesis: your conclusion from the evidence, given in an oral or written report.

PART 1 · WORKED EXAMPLE

Superior Bicycles

A textbook case: the five steps applied to suspected embezzlement

THE CASE

An audit finds unaccounted spending in accounts payable. The accountant, Jamie Poole, is making many luxury purchases.

QUESTION

How do the five steps apply?

1	**Hypothesis**	The purchases are not job-related.
2	**Artifacts**	His laptop and smartphone, and the accounts server he can access.
3	**Collect**	Acquire them with proper evidence procedures.
4	**Analyse**	Test the hypothesis. If evidence disputes it, revise and repeat.
5	**Thesis**	Report the findings and opinions.

PART 1 · WORKED EXAMPLE

When the Hypothesis Shifts

A textbook case: George Montgomery, a sales representative

	WHAT IS KNOWN	HYPOTHESIS	EVIDENCE SOUGHT
First report	Complaints about his work; absent two days, as is a colleague	**Something explains his absence and poor work**	Clues on his drive and storage media
After talking to coworkers	He runs a side business on company computers	**He is misusing company resources**	Websites, web addresses and internet providers on his USB drive

Stay objective: he is suspected only of misuse, and the evidence may clear him.

PART 2 OF 3

A Systematic Plan

Planning a case so that nothing is missed

02

IN THIS PART YOU WILL LEARN TO

- List the tasks of a systematic approach
- Assess a case before planning it

KEY TERMS

Initial assessment **Risk** **Contingency plan**

PART 2 · METHOD

Planning a Case: Tasks 1 to 6

From the first conversation to knowing what could go wrong

1. **Make an initial assessment**
   Ask what kind of case it is and what was seized.
2. **Choose a general approach**
   Outline the main steps, such as when to seize a computer.
3. **Write a detailed checklist**
   Each step, with an estimated time for it.
4. **List the resources**
   Software, tools and expert help suited to the system.
5. **Obtain and copy the evidence**
   Make a forensic image (Lesson 0.3) of each drive.
6. **Identify the risks**
   Such as a password trap that wipes the drive.

Scale the plan: simple for most cases; detailed, and reviewed often, for complex ones.

PART 2 · METHOD

Planning a Case: Tasks 7 to 12

From reducing risk to reviewing the finished case

7 Reduce the risks

For example, make several copies of the original media.

8 Test the design

Review decisions; compare hash values to check copies.

9 Analyse and recover

Examine the copy with your tools to find evidence.

10 Investigate what you recover

Review files, deleted files, email and web history.

11 Complete the case report

Write up what you did and what you found.

12 Critique the case

Review what went well, and what you would do better.

Contingency plan: always have other tools or methods ready in case one fails.

PART 2 · WORKED EXAMPLE

Assessing the Montgomery Case

A textbook case: the assessment that shapes the plan

CASE ASSESSMENT · MONTGOMERY_72022

SITUATION	**Misuse of company resources**
NATURE	**A side business at work**
EVIDENCE	**One small USB drive**
DISK FORMAT	**NTFS (Lesson 0.2)**
LOCATION	**His assigned computer**
POLICY	**Assets open to inspection**

WHAT THE ASSESSMENT TELLS YOU

What to look for

Websites, web addresses and internet providers.

What you need

A reliable tool to copy the drive and find deleted files.

What you can skip

Seizing the drive: the IT department already has it.

The task: confirm or deny that he ran a side business on company time.

PART 3 OF 3

Conduct and Notes

Keeping your work credible from the first day

03

IN THIS PART YOU WILL LEARN TO

- Explain what professional conduct means
- Write a first examiner-notes entry

KEY TERMS

Objectivity **Confidentiality** **Examiner notes**

PART 3 · STRUCTURE

Professional Conduct

What your credibility as an examiner rests on

YOUR CREDIBILITY

Objectivity

Base opinions on the evidence, your training and experience. Avoid bias.

Confidentiality

Discuss the case only with people who need to know.

Keep learning

Training, conferences, membership and study keep you current.

Integrity: careless conduct anywhere in your life can be used to discredit you in court.

PART 3 · CONTRAST

Conduct in Practice

Habits to follow, and habits to avoid

DO

- Follow every reasonable lead before reaching a conclusion
- Share only general facts when asking others for advice
- Keep the case confidential until a court or lawyer needs a report

AVOID

- Letting a client's or lawyer's agenda decide the outcome
- Naming people or details to anyone outside the case
- Discussing a dismissed employee: the company could be sued

Attorney-work-product rule: when you work for a lawyer, discuss the case only with the lawyer's team, unless the lawyer approves.

PART 3 · CONCEPT

Examiner Notes

A journal of the examination, started on day one

What they are

A record, updated for every task, of what you did and what you found.

Why they matter

They support the report, and let another examiner repeat your steps.

What to record

The date and time, the task, and specific findings, stated as facts.

⚠ **Your notes can be read in court, and by the other side.** Write only facts, keep opinions out, and store the notes securely.

PART 3 · WORKED EXAMPLE

The First Notes Entry

The Montgomery case: one entry in a notes log, and what each column records

1 · DATE AND TIME	2 · CASE	3 · ITEM AND SOURCE	4 · ACTION AND RESULT
20 July 2022, 10:05	`Montgomery_72022`	USB drive, item 1, from the IT manager	Sealed in an evidence bag, logged on the form, locked in the evidence cabinet

1 When

The date, and the time the task started.

2 Which case

The case name or number.

3 Item and source

The item, and who handed it over.

4 What was done

Each action and its result, as plain fact.

An illustrative entry. Later lessons add hash values and tool details.

LESSON 1.3 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Five steps of an investigation

Hypothesis, artifacts, collect, analyse, then a thesis. Steps 1 to 4 repeat as needed.

2

A systematic plan

Twelve tasks, from initial assessment to critique, with risks and a contingency plan.

3

Conduct and notes

Objective, confidential and current, with notes that record every task as fact.

Work to a plan, and write down everything you do as you do it.

UP NEXT

Lesson 1 Summary

Next, a short review brings Lessons 1.1 to 1.3 together before Lesson 2.



What digital forensics is



Who may investigate



How to approach a case

DFBOOK DIGITAL FORENSICS · LESSON 1 · SUMMARY

Digital Forensics and the Investigator: Review

The field, the authority and the method, in one short review



IN THIS LESSON

1. What digital forensics is
2. Public and private investigations
3. A systematic approach to a case

LESSON 1 · REVIEW

Lesson 1 at a Glance

One column for each sub-lesson

1.1 The field

1. Computer science and investigative procedure, applied to digital evidence.
2. Its goal sets it apart from data recovery and incident response.
3. Look for inculpatory and exculpatory evidence alike.

1.2 The authority

1. Public cases rely on legal powers, such as a search warrant.
2. Companies rely on policies, banners and an authorised requester.
3. Some crimes, such as illegal images, must go to the police.

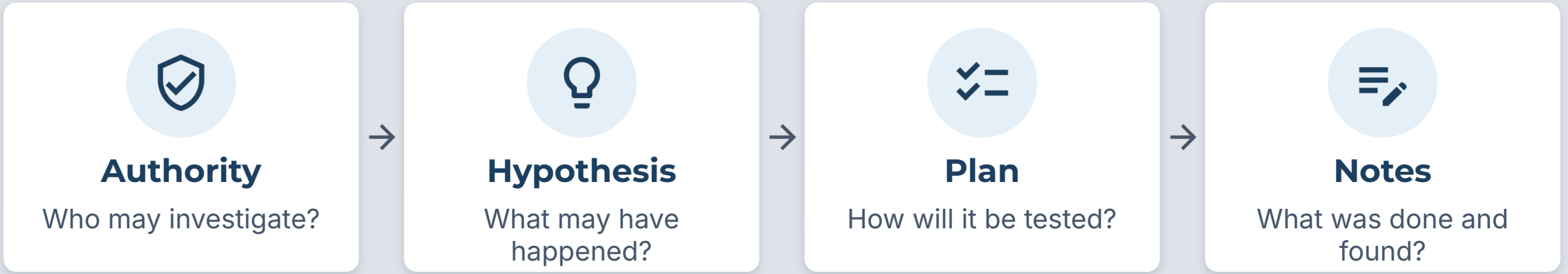
1.3 The method

1. Five steps, from a hypothesis to a thesis.
2. A plan of twelve tasks, including risks.
3. Objective, confidential conduct, and examiner notes.

LESSON 1 · REVIEW

From Request to Report

How the three sub-lessons fit together on a case



Confirm your authority, test a hypothesis to a plan, and record every step.

UP NEXT

Handling Digital Evidence at the Scene

Lesson 2 covers what beginners are most likely to do first: recognise, protect and record evidence.

- **Identifying digital evidence**
- **Preparing and securing the scene**
- **Bagging, tagging and custody**

